

MERN-BASED ONLINE LEARNING PLATFORM

A PROJECT REPORT

*submitted in partial fulfilment of the requirements for the
degree of*

Bachelor of Technology

in

COMPUTER ENGINEERING

Major Project I (01CE0716)

Submitted by

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Marwadi University, Rajkot

August, 2025



Major Project I (01CE0716)

Department of Computer Engineering

Faculty of Engineering & Technology

Marwadi University

A.Y. 2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **MERN-BASED ONLINE LEARNING PLATFORM** has been carried out by Anurag Tiwari (92201703100), under my guidance in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering, 7th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Prof. Rupesh Kanojiya

Assistant Professor

Department of Computer Engineering

Prof. (Dr.) Krunal Vaghela

Professor & Head

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DECLARATION

We hereby declare that the **Major Project-I (01CE0716)** report submitted along with the Project entitled **MERN-BASED ONLINE LEARNING PLATFORM** submitted in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering to Marwadi University, Rajkot, is a Bonafide record of original project work carried out by me / us at Marwadi University under the supervision of **Prof. Rupesh Kanojiya** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

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Acknowledgement

We would like to express our sincere gratitude to our internal guide, Prof. Rupesh Kanojiya, for his continuous support, guidance, and encouragement throughout the development of our project titled “**MERN-based online learning platform : EdTech**”. His valuable suggestions and feedback helped us improve our project and learn new skills.

We also thank our college and faculty members for providing us with the opportunity and necessary resources to work on this project. I also extend my sincere appreciation to Dr. Krunal Vaghela (HOD) and other faculty members for their encouragement and support.

I am grateful to my classmates and friends for their motivation and helpful discussions. Lastly, we acknowledge the authors and online sources that served as references for this work. This project would not have been possible without the teamwork and dedication of every member.

Thank you.

Anurag Tiwari

MERN-based online learning platform : EdTech

Marwadi University

Abstract

This project presents the design and development of a fully functional MERN-based online learning platform : EdTech using the MERN stack (MongoDB, ExpressJS, ReactJS, NodeJS). The aim of the project is to create a user-friendly, secure, and scalable platform where students can discover, purchase, and take online courses instructors can create and manage courses while tracking student engagement, and admins can manage users and monitor platform performance. The application provides a dynamic and responsive interface, ensuring a smooth user experience across devices. MongoDB handles backend data storage, with ExpressJS and NodeJS managing serverside logic, and ReactJS with Tailwind CSS and Redux for the front-end. The project uses JWT (JSON Web Token) authentication, which securely validates user identity and ensures that only authorized users can access and perform actions such as course creation, enrollment, or analytics viewing. Key features include user registration and login, course discovery and purchase, course creation and management, progress tracking, payment integration via Razorpay , instructor dashboards, real-time analytics, notifications, and an admin panel for platform oversight. This project demonstrates the power of the MERN stack in building secure, full-stack EdTech applications, and provides a foundation for future enhancements such as AI-based recommendations and mobile app integration. suggestions and cloud deployment.

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Abbreviations

MERN	: MongoDB , ExpressJS ,ReactJS ,NodeJS
JWT	: JSON Web Token
UI	: User Interface
UX	: User Experience
API	: Application Programming Interface
DB	: Database
CRUD	: Create, Read, Update, Delete
VS Code	: Visual Studio Code

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CHAPTER 1

INTRODUCTION

StudyNotion is a modern, user-friendly **EdTech Learning Management System** web application designed to simplify online education. It enables students to discover, purchase, and take courses, instructors to create and manage courses while tracking engagement, and admins to manage users and monitor performance. The platform supports secure authentication, responsive design, and a dedicated admin panel for efficient user and content management. By integrating key features like course enrollment, payments, notifications, and analytics, StudyNotion offers a seamless and convenient experience for educational users.

1.1.1 Need for an EdTech LMS

With the rapid growth of online education, students and instructors need tools to discover, deliver, and manage courses efficiently. An EdTech LMS bridges this gap by providing a unified platform for course creation, enrollment, progress tracking, and analytics, saving time and improving learning outcomes.

1.1.2 What is a LMS?

LMS is a software application that allows students to browse and enroll in courses, instructors to upload and manage educational content, and admins to oversee the platform. It provides essential features such as course discovery, payment processing, progress tracking, instructor analytics, and notifications. Popular examples include Coursera, Udemy, and edX. These platforms handle both the frontend user experience and backend data management.

1.1.3 Tools and Technologies Used

This project uses the MERN stack (MongoDB, ExpressJS, ReactJS, NodeJS) for building a full-stack web application. Additional tools and libraries include:

- **JWT Authentication** for secure user login
- **Razorpay** for payment integration
- **Tailwind CSS & Redux** for styling and state management
- **Cloudinary** for media hosting (images, videos, documents)

1.1.4 Features of the System

Highlighting the main functionalities your EdTech LMS provides:

- User registration and login
- Course discovery, purchase, and enrollment
- Course creation, editing, and management for instructors
- Progress tracking and completion certificates

- Admin panel for managing users and courses
- JWT authentication with role-based access control

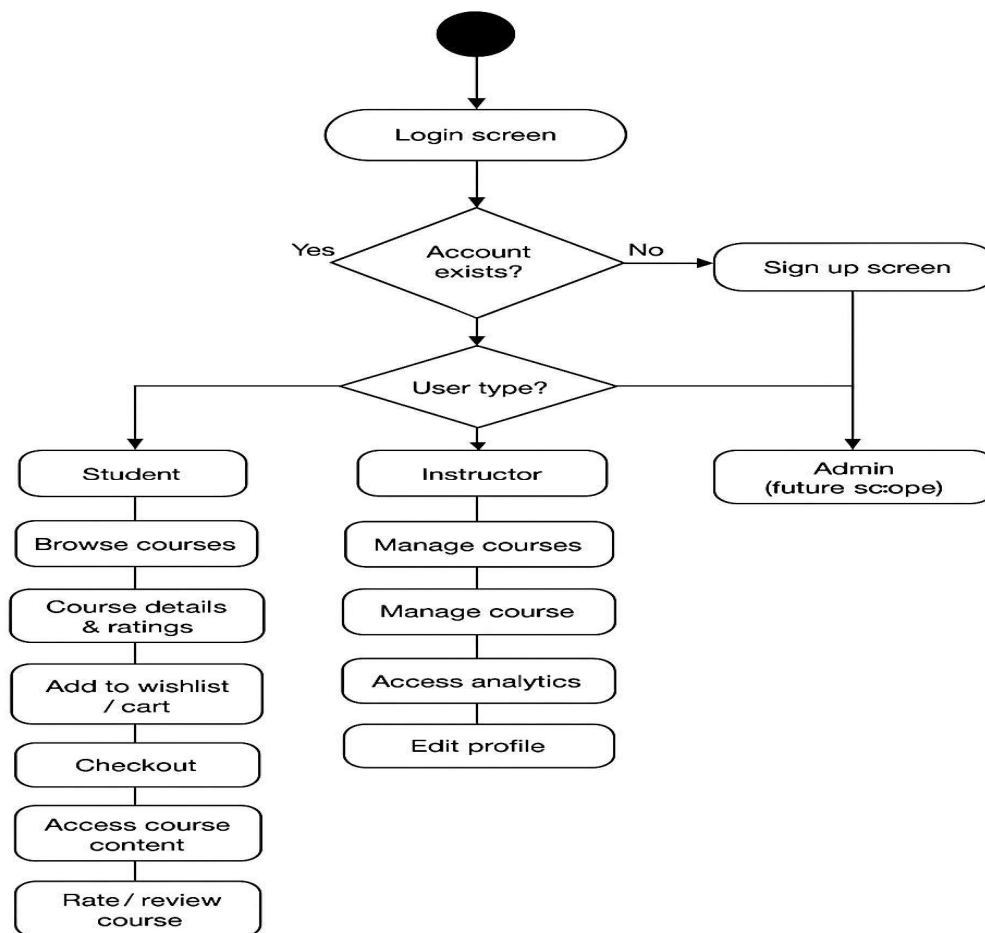


Figure: 1.1 Work Flow Diagram

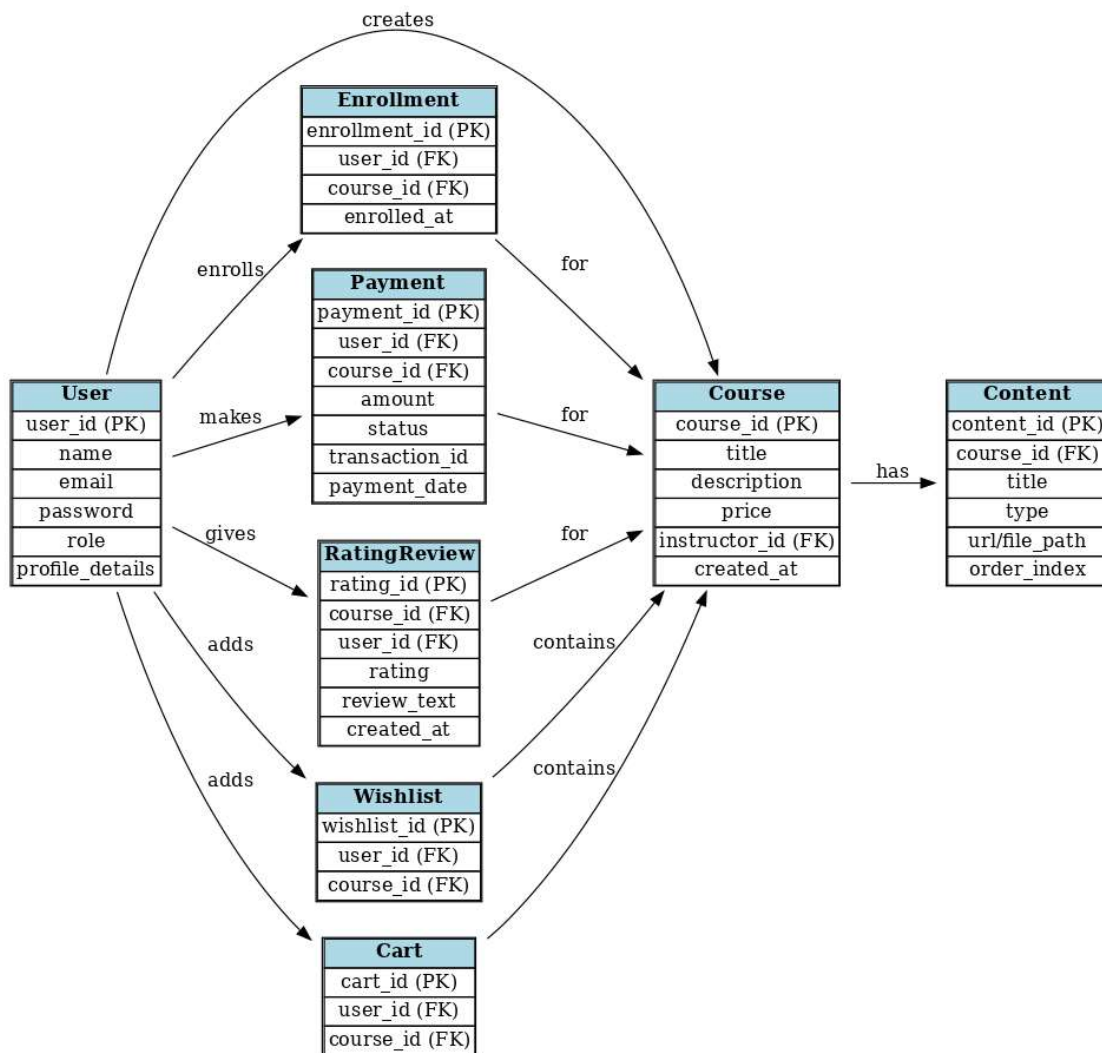


Figure: 1.2 Entity-Relationship Diagram

CHAPTER 2

TECHNOLOGY STACK & SYSTEM ARCHITECTURE

The StudyNotion EdTech Platform is built using the MERN stack for full-stack web development. The architecture ensures scalability, real-time updates, and a secure environment for handling course interactions such as enrollment, progress tracking, and analytics. This chapter explains the technologies used and how the system architecture is designed .

2.1 Technology Stack

2.1.1 React Js(Frontend Framework)

ReactJS is a popular JavaScript library used to build interactive user interfaces for web applications.

- Provides a **responsive UI** with reusable components.
- Supports **React Router** for seamless navigation across screens.
- Styled with **TailwindCSS** for a modern and adaptive user interface.
- Uses **Redux** for global state management, handling user sessions and course data.

For Instructors:

- **Dashboard:** This page will have an overview of the instructor's courses, as well as the ratings and feedback for each course.
- **Insights:** This page will have detailed insights into the instructor's courses, including the number of views, clicks, and other relevant metrics.
- **Course Management Pages:** These pages will allow the instructor to create, update, and delete courses, as well as manage the course content and pricing.
- **View and Edit Profile Details:** These pages will allow the instructor to view and edit their account details.

For Admin (this is for future scope):

- **Dashboard:** This page will have an overview of the platform's courses, instructors, and students.
- **Insights:** This page will have detailed insights into the platform's metrics, including the number of registered users, courses, and revenue.
- **Instructor Management:** This page will allow the admin to manage the platform's instructors, including their account details, courses, and ratings. Admin will also have the ability to remove instructor ratings if required.

Mentorship Module (Future): Once implemented, this module will allow users to follow learning paths, interact with mentors, and receive recommendations. Admin will oversee this module to ensure smooth functioning.

For Students:

- Homepage: This page will have a brief introduction to the platform, as well as links to the course list and user details.
- Course List: This page will have a list of all the courses available on the platform, along with their descriptions and ratings.
- Wishlist: This page will display all the courses that a student has added to their wishlist.
- Cart Checkout: This page will allow the user to complete the course purchase.
- Course Content: This page will have the course content for a particular course, including videos, and other related material.
- User Details: This page will have details about the student's account, including their name, email, and other relevant information.
- User Edit Details: This page will allow the student to edit their account details.

2.1.2 ExpressJS & NodeJS (Backend)

NodeJS is a runtime for server-side JavaScript, with ExpressJS as a framework for building APIs.

- Manages user data, courses, enrollments, and payments.
- Provides RESTful APIs for CRUD operations.
- Integrates with Razorpay for secure payments and Cloudinary for media hosting.
- Handles real-time features using WebSockets (optional via Socket.io).

2.1.3 MongoDB (Database)

MongoDB is a NoSQL database for flexible data storage.

- Stores user profiles, courses, enrollments, and analytics in schemas.
- Supports efficient querying for course search and progress tracking.
- Integrates easily with NodeJS using Mongoose for schema definition

2.1.4 Supporting Tools

Additional tools used in development include:

- **JWT** → Secure authentication.
- **Razorpay** → Payment gateway.
- **Cloudinary** → Media storage for course videos and documents.
- **VS Code** → Development environment.
- **Postman** → API testing.
- **Bcrypt** → for password hashing, adding an extra layer of security.
- **Git & GitHub** → Version control and collaboration.

2.2 System Architecture

The StudyNotion follows a client-server architecture with three main layers:

2.2.1 Frontend (Client Side)

The client is built using React JS. It provides:

- User authentication pages (login, register).
- Main features (course catalog, enrollment, instructor course creation).
- Dashboards for students (progress) and instructors (analytics).
- Notifications and payment checkout.
- Responsive design using TailwindCSS.

2.2.2 Backend (Server Side)

Supabase acts as the backend, managing:

- Authentication & Authorization with JWT.
- API endpoints for CRUD operations on posts, likes, comments, and accounts.
- Business logic such as post scheduling and analytics tracking.
- Admin panel APIs for managing users and content.

2.2.4 Data Flow Summary

The flow of data in the system follows this pattern:

1. **Client (ReactJS Frontend)** sends a request (e.g., user logs in, views courses, or purchases a course).
2. **Backend (Node.js + Express.js)** receives the request, validates it using **JWT authentication**, and processes the business logic.
3. **Database (MongoDB Atlas)** is queried or updated with the required information (e.g., user details, course data, purchase records).
4. **Cloudinary (Media Management)** is accessed if the request involves course media (videos, images, PDFs).
5. The **Backend** sends a structured response back to the **Frontend**, ensuring smooth interaction (e.g., displaying updated course details or successful enrollment).

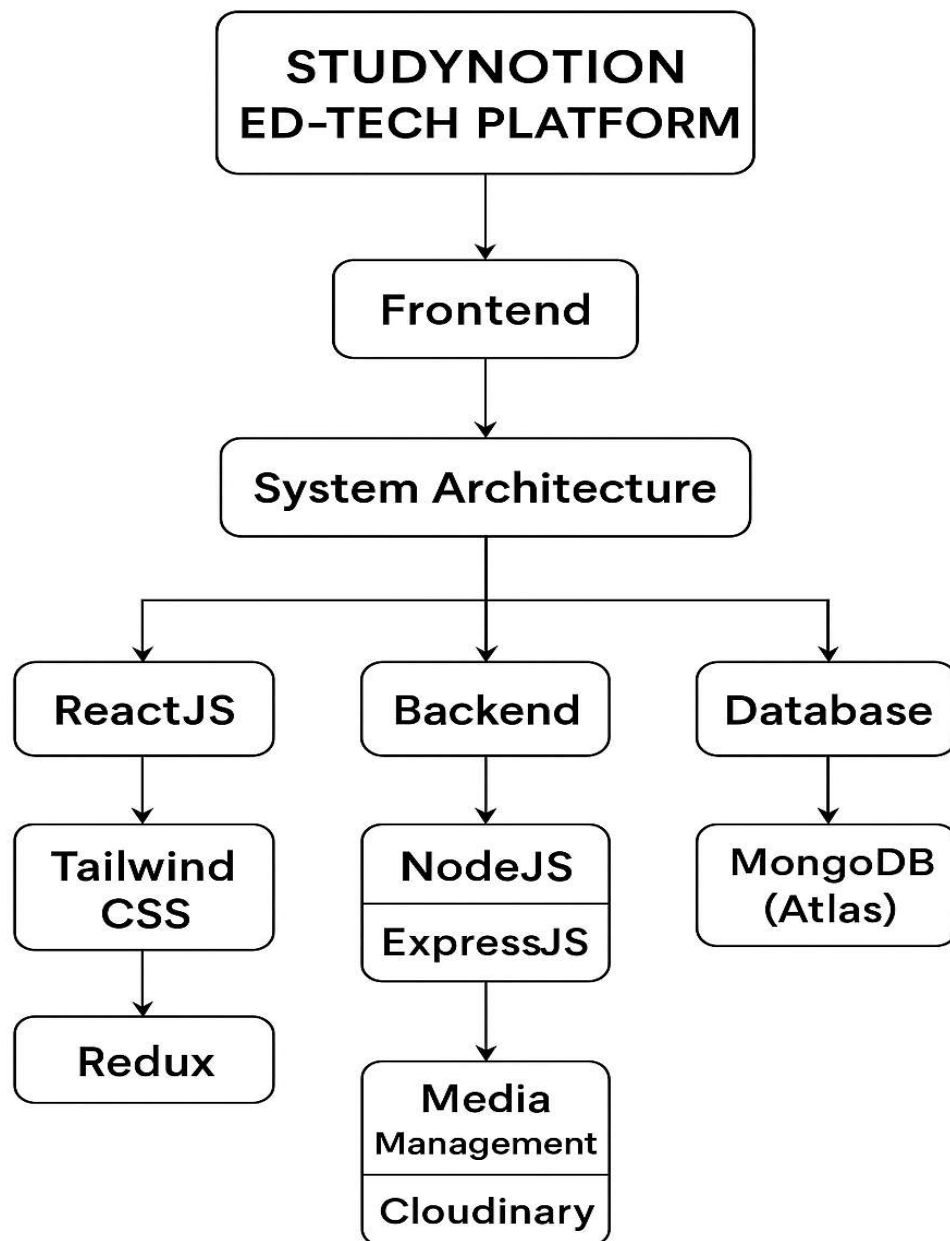


Figure: 2.1 Layered Architecture Diagram

CHAPTER3

MODULES OF THE SYSTEM

The StudyNotion EdTech Platform is divided into distinct modules, each responsible for a specific set of functionalities. Modular design simplifies development, testing, maintenance, and understanding of the system. The following are the main modules of the EdTech LMS:

3.1 User Management Module

This module manages all **user-related operations**:

- **Registration & Login:** Users can create accounts and securely log in using JWT.
- **Profile Management:** Users can update personal information like profile picture, name, and bio.
- **Role-Based Access Control:** Differentiates between students, instructors, and admins.
- **Security:** Passwords are securely hashed, and sessions are managed with tokens.

3.2 Course Management Module

Handles all operations related to courses:

- **Create, Edit, Delete Courses:** Instructors can manage courses efficiently.
- **Content Attributes:** Each course includes sections, lectures, videos, documents, quizzes, and metadata.
- **Visibility & Pricing:** Instructors can set prices, thumbnails, and tags for discoverability.
- **Database Integration:** Changes are stored in MongoDB and reflected via API calls.

3.3 Enrollment & Progress Module

Manages user engagement on posts:

Manages student engagement with courses:

- Students can enroll via **payment**.
- Track progress, mark lectures as completed, and issue certificates.
- Engagement is tracked for analytics.
- Notifications are sent for course updates or completions.

3.4 Instructor Dashboard Module

Allows instructors to monitor and manage their courses :

- **View student enrollments and progress.**
- **Analytics on course performance (completion rates, reviews).**
- **Upload new content and edit existing courses.**

3.5 Payment & Notification Module

Enables secure transactions and updates:

- Integrate Razorpay for course purchases.
- **Push notifications for enrollments, course updates, or payments.**
- Real-time updates ensure users see the latest course changes.

3.6 Admin Panel Module

A dedicated module for **administrative tasks**:

- Admins can **manage users, posts, and comments.**
- Provides dashboards for **analytics, reports, and user activity monitoring.**
- Supports CRUD operations and role management.
- Ensures secure access restricted to authorized personnel only.

3.7 Analytics & Insights Module

Provides detailed **user engagement analytics**:

- Tracks course **views, enrollments, completions, and reviews.**
- Provides **metrics** such as **retention rate, revenue, and student feedback.**
- Helps **instructors and admins understand user behavior and content** performance.

3.8 Real-Time Updates Module

Ensures all activities are synchronized across devices instantly:

- New course updates, enrollments, and progress are updated without reloads.
- Implemented using Redux and API polling.
- Provides a seamless and interactive user experience.

3.9 Notification Module

Keeps users informed of important events:

- Alerts for new enrollments, course updates, payments, or progress milestones.
- Push notifications delivered in-app and optionally via email.
- Helps users stay engaged and respond promptly to changes.

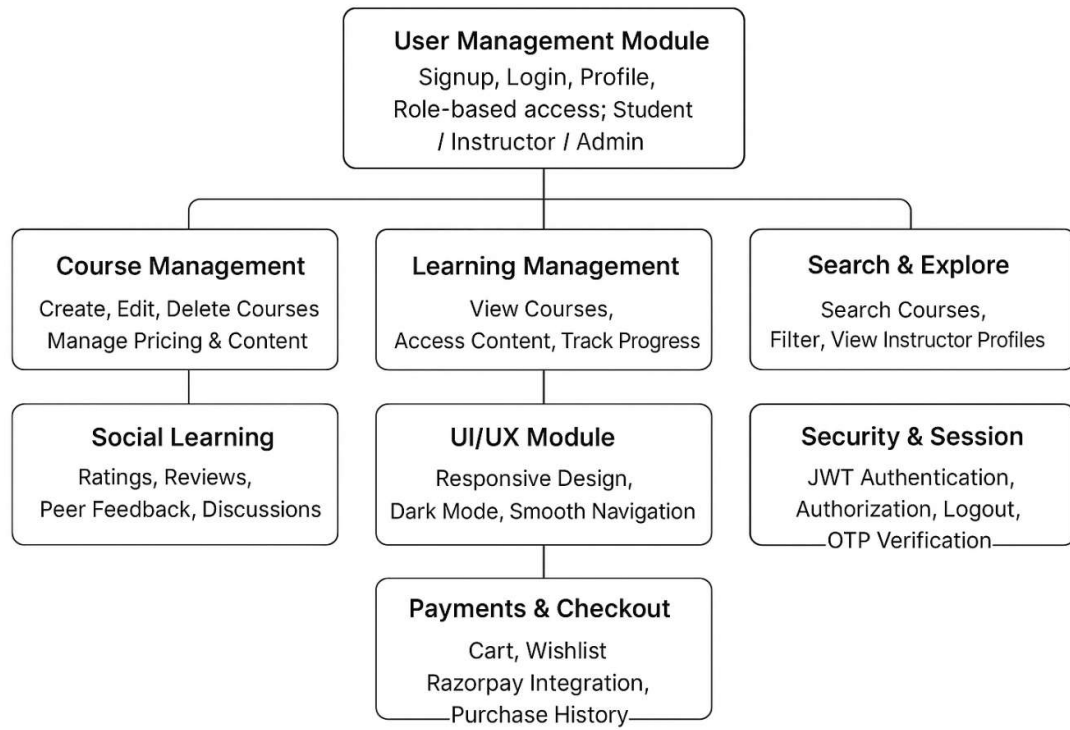


Figure 3.1: System Modules Diagram

Snapshots of each module UI (Login Page, Home Page, Collections, Admin Panel, etc.)

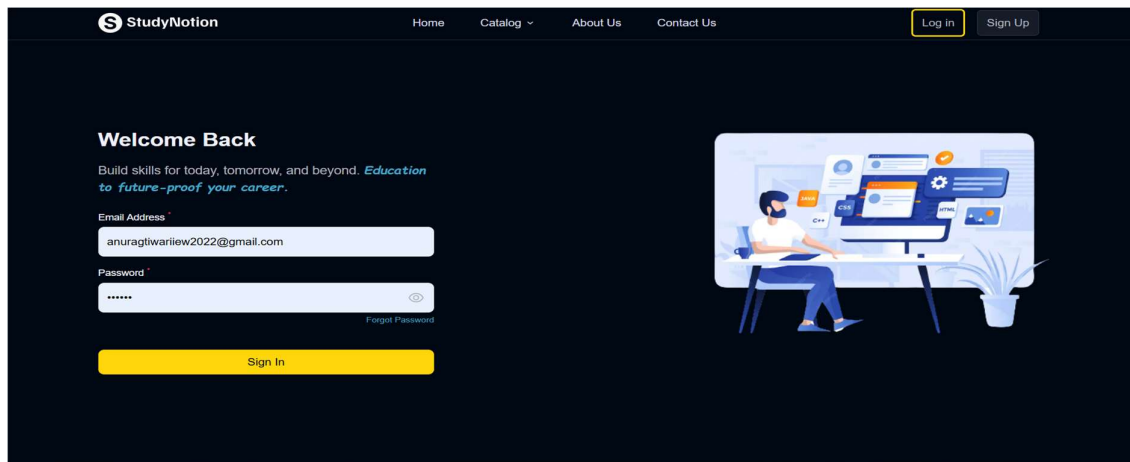


Figure: 3.2 Login Page

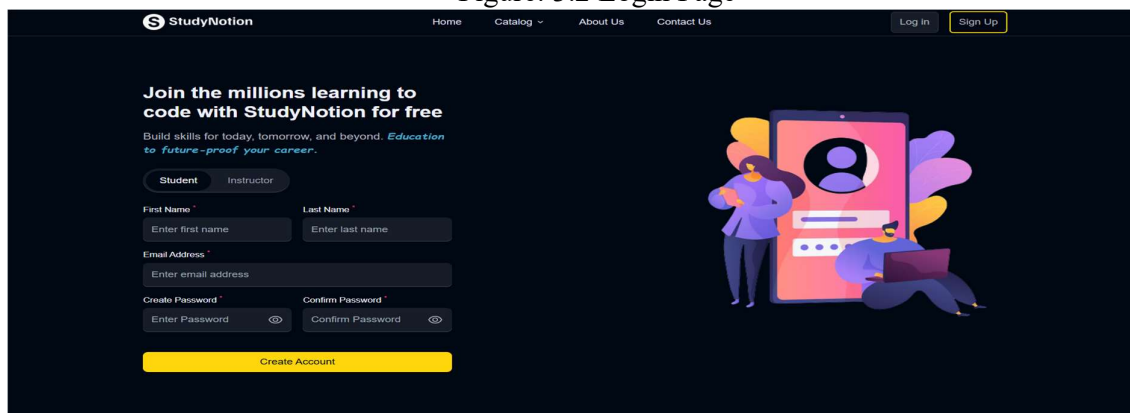


Figure: 3.3 Signup Page

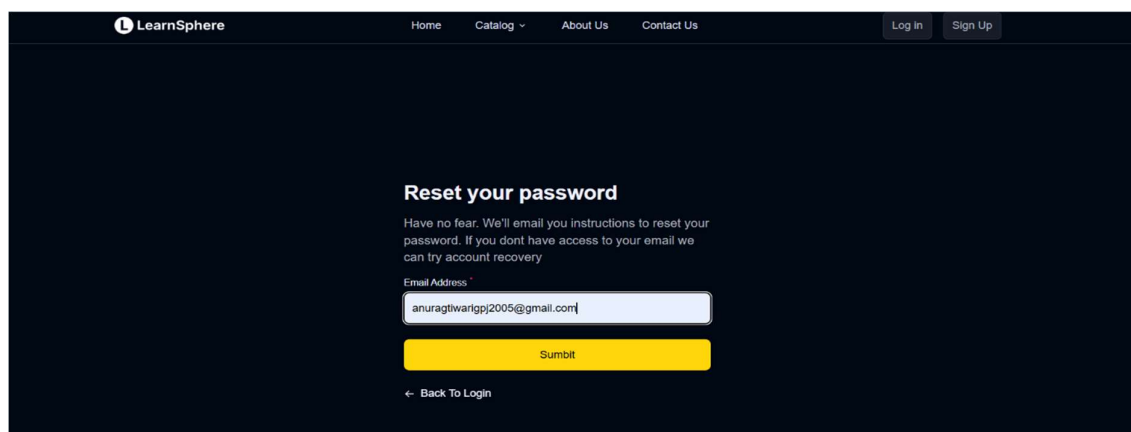
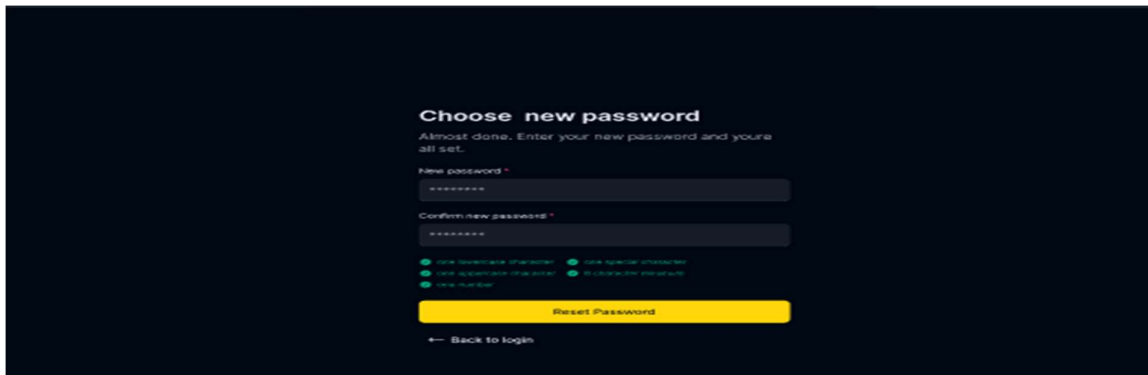
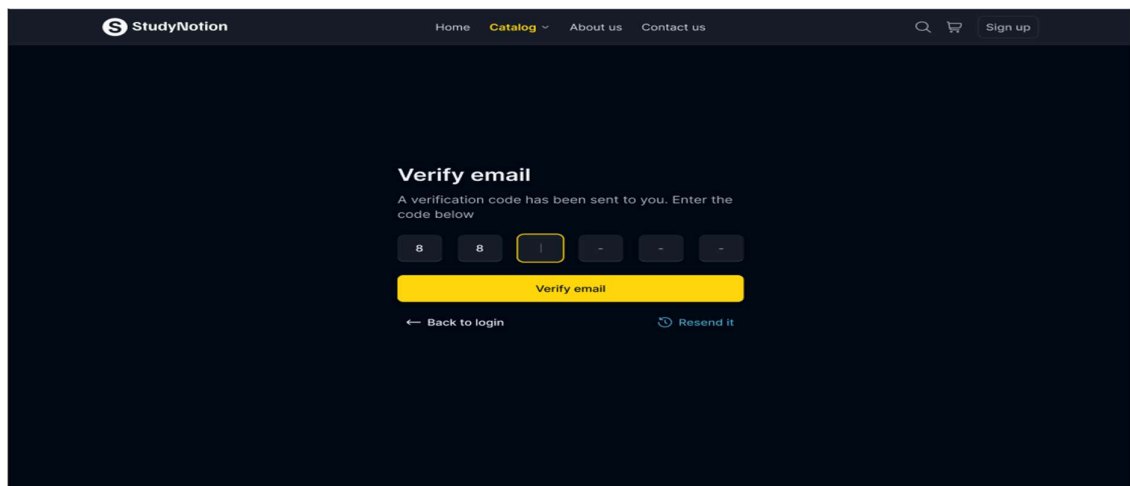


Figure: 3.4 Reset Password Page



The image shows a 'Choose new password' page with a dark background. It includes two input fields for 'New password' and 'Confirm new password', both masked with dots. Below the fields are four green checkmarks indicating password requirements: one uppercase character, one lowercase character, one numeric character, and one special character. A yellow 'Reset Password' button is at the bottom, along with a link to 'Back to login'.

Figure: 3.3 Reset Password Page



The image shows a 'Verify email' page for StudyNotion. It features a navigation bar with 'Home', 'Catalog', 'About us', and 'Contact us'. The main content area has a title 'Verify email' and a message: 'A verification code has been sent to you. Enter the code below'. There are five input boxes for the code, with the first two containing the digit '8'. A yellow 'Verify email' button is below the boxes. At the bottom, there are links for 'Back to login' and 'Resend it'.

Figure: 3.4 OTP Verify Page

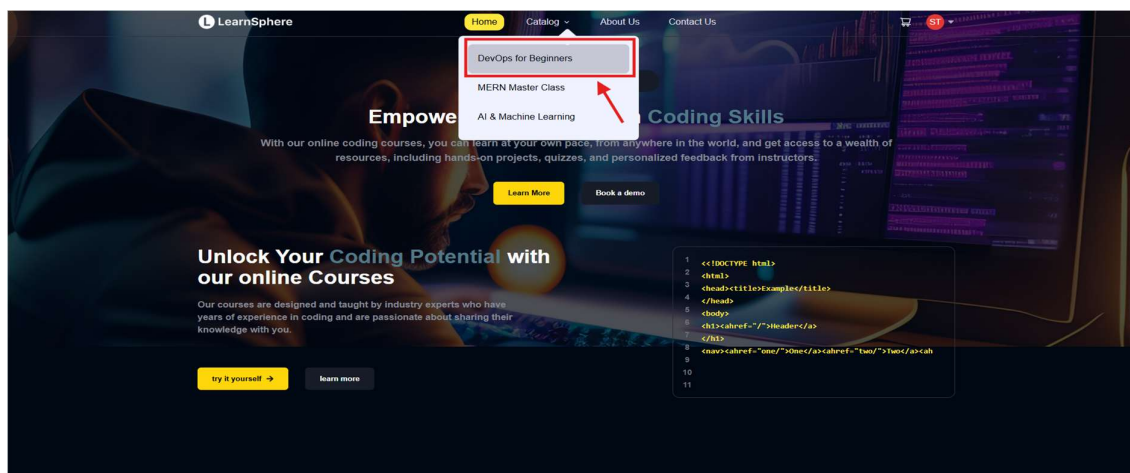


Figure: 3.5 Course Catalog Page

The screenshot displays the LearnSphere website interface. At the top, the navigation bar includes 'Home', 'Catalog', 'About Us', and 'Contact Us'. The main header area features the 'DevOps for Beginners' course title and a sub-header 'Learn Web Dev with industry experts, project based learning'. Below this, a section titled 'Courses to get you started' showcases three featured courses: 'Java Masterclass' (Rs. 4999), 'React.js Zero to Hero' (Rs. 3999), and 'Spring & Spring Boot Zero to Hero' (Rs. 6999). Each course card includes a rating and a 'Buy Now' button.

The 'Frequently Bought' section displays four courses: 'Java Masterclass' (Rs. 4999), 'React.js Zero to Hero' (Rs. 3999), 'Spring & Spring Boot Zero to Hero' (Rs. 6999), and 'HTML, CSS & JavaScript Fundamentals of Development' (Rs. 1999). Each course card includes a rating and a 'Buy Now' button.

The 'Java Masterclass' course detail page is shown below, featuring a 'What you'll learn' section with four bullet points: 'Master DevOps practices for Java projects', 'Build CI/CD pipelines to automate Java app delivery', 'Containerize Java applications using Docker', and 'Deploy and scale Java apps seamlessly on Kubernetes'. The page also includes a 'Course Requirements' section, a 'Tags' section (Java, Backend, Spring Boot), and an 'Author' section (Vladimir Putin).

Java Masterclass
Completed Java course from basics to advanced
0 ★★★★★ (0 reviews) undefined students enrolled
Created by Vladimir Putin
Created at August 25, 2025 @ English

What you'll learn

- Master DevOps practices for Java projects
- Build CI/CD pipelines to automate Java app delivery
- Containerize Java applications using Docker
- Deploy and scale Java apps seamlessly on Kubernetes

Course Requirements :

- Basic Java knowledge
- Containerization
- Subnet configuration

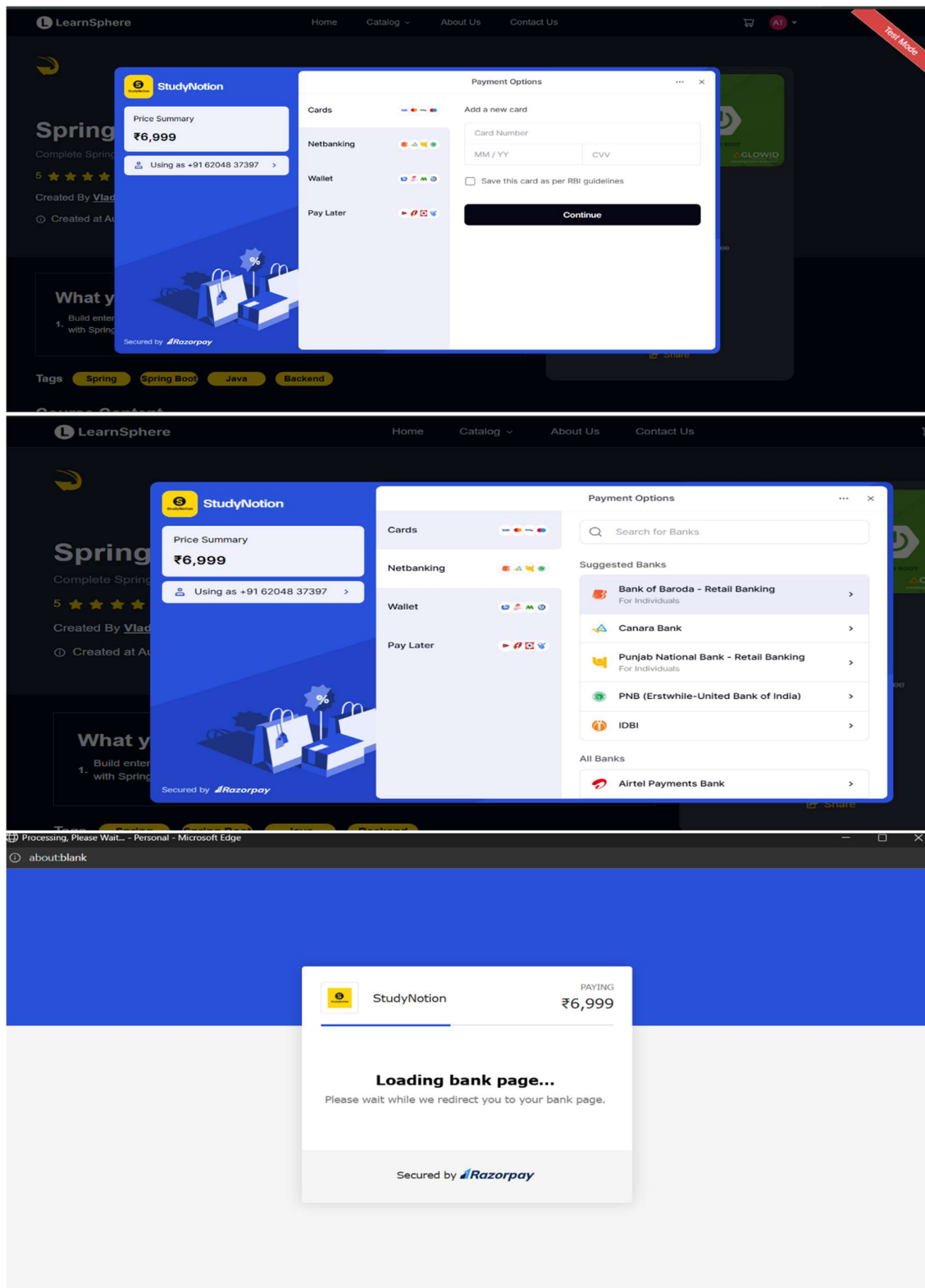
Tags Java Backend Spring Boot

Course Content
0 sections() 0 lectures() 0s Total Time

Author
VP Vladimir Putin

Rs. 4999
Buy Now
Add to Cart
30 Day Money-Back Guarantee

Figure: 3.6 Course Detail Pages



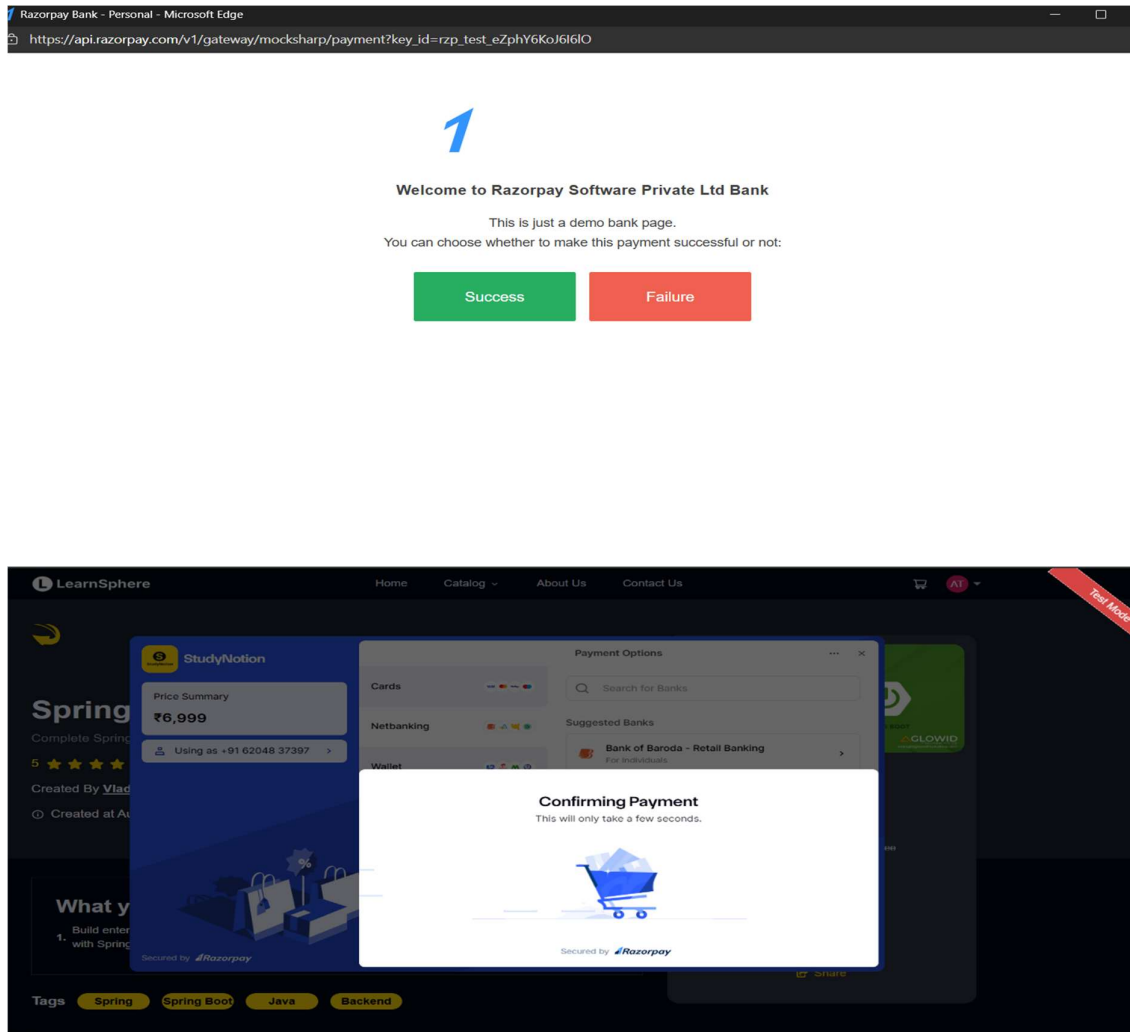


Figure: 3.7 Razorpay Payment Images

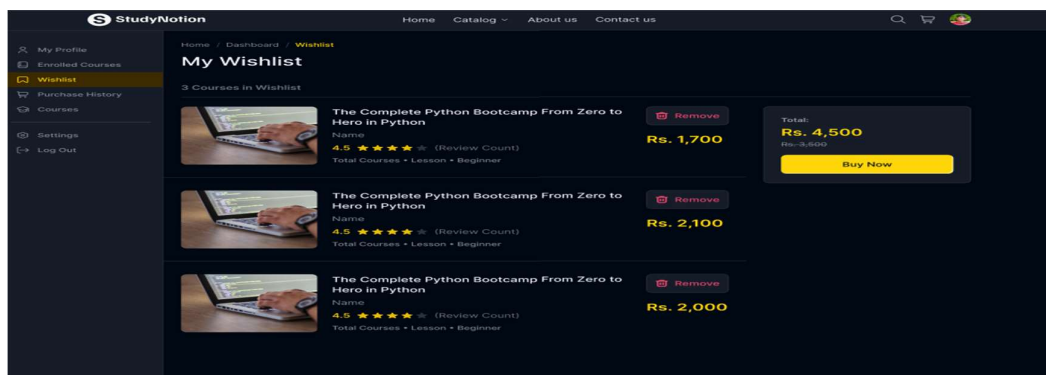


Fig 3.8 Wishlist Page

StudyNotion Home Catalog About us Contact us

Home / Learning / **Python**

The Complete Python Bootcamp From Zero to Hero in Python

This Python for beginners course will help you to become Zero to Hero. Learn Python Programming in Easy Way.

4.5 ★★★★★ (650 ratings) 332,402 students

Created by Instructor Name

🕒 Created at 02/2020 🌐 English

What you'll learn

- Introduction to Python and Python 3
- Understand the basics: Data types, Loops, Conditional statements, Functions and Modules
- Learn object oriented programming in Python
- Learn how to make your own web-scraping tool using Python
- Know how to Read and Parse JSON and XML files

Course content

10 sections • 41 lectures • 7h 57m total length

Introduction & Basics 5 lectures • 51min

- Why Python 02:09
- How to Install Python and Sublime text 02:09
- Variable Declaration and Memory Allocation 02:09
- Python Data Types 02:09
- Python Operators 02:09
- Conditional Statements and Loops 5 lectures • 51min
- Python Data Types - String, Lists, Tuple, Dictionaries 5 lectures • 51min
- Python Functions 5 lectures • 51min
- Python Modules and Packages 5 lectures • 51min

Author

Rs. 1,200

Add to Cart

Buy now

30-Day Money-Back Guarantee

This course includes:

- 8 hours on-demand video
- Full Lifetime access
- Access on Mobile and TV
- Certificate of completion

Share

Fig 3.9 Video Detail Page

Unlock Your Coding Potential with our online Courses

Our courses are designed and taught by industry experts who have years of experience in coding and are passionate about sharing their knowledge with you.

try it yourself **Learn more**

```
1 <!DOCTYPE html>
2 <html>
3 <head><title>Example</title>
4 </head>
5 <body>
6 <h1>Hello World</h1>
7 </body>
8 </html>
```

Unlock Your Coding Potential with our online Courses

Our courses are designed and taught by industry experts who have years of experience in coding and are passionate about sharing their knowledge with you.

try it yourself **Learn more**

Unlock the Power of Code

Learn to Build Anything You Can Imagine

Free **How to coding** **Most popular** **Web path** **Course path**

Learn HTML

This course covers the basic concepts of HTML, including creating and structuring web pages, adding text, links, images, and more.

Beginner **5 Lessons**

Learn CSS

This course provides a detailed guide to CSS, covering selectors, styling elements, layout, and responsive design techniques.

Beginner **9 Lessons**

Responsive Web design

This course focuses on creating web designs that are responsive and adapt to different devices and screen sizes.

Beginner **11 Lessons**

Explore Full Coding **Learn more**

Build the skills that top employers are looking for today

The world's leading companies are looking for people who can build the skills that top employers are looking for today. Today, to be a competitive specialist requires more than professional skills.

Learn more

Fig 3.10 Hero Section on landing Page



Become an Instructor

Instructors from around the world teach millions of students on StudyNotion. We provide the tools and skills to teach what you love.

[Start Learning Today →](#)

Reviews from other learners


[Home](#)
[Catalog](#)
[About Us](#)
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[Log in](#)
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Driving Innovation in Online Education for a Brighter Future

StudyNotion is at the forefront of driving innovation in online education. We're passionate about creating a brighter future by offering cutting-edge courses, leveraging emerging technologies, and nurturing a vibrant learning community.



We are passionate about revolutionizing the way we learn. Our innovative platform combines technology, expertise, and community to create an unparalleled educational experience.

Our Founding Story

Our e-learning platform was born out of a shared vision and passion for transforming education. It all began with a group of educators, technologists, and lifelong learners who recognized the need for accessible, flexible, and high-quality learning opportunities in a rapidly evolving digital world.

As experienced educators ourselves, we witnessed firsthand the limitations and challenges of traditional education systems. We believed that education should not be confined to the walls of a classroom or restricted by geographical boundaries. We envisioned a platform that could bridge these gaps and empower individuals from all walks of life to unlock their full potential.



Our Vision

With this vision in mind, we set out on a journey to create an e-learning platform that would revolutionize the way people learn. Our team of dedicated experts worked tirelessly to develop a robust and intuitive platform that combines cutting-edge technology with engaging content, fostering a dynamic and interactive learning experience.

Our Mission

Our mission goes beyond just delivering courses online. We wanted to create a vibrant community of learners, where individuals can connect, collaborate, and learn from one another. We believe that knowledge thrives in an environment of sharing and dialogue, and we foster this spirit of collaboration through forums, live sessions, and networking opportunities.

5K

Active Students

10+

Mentors

200+

Courses

50+

Awards

World-Class Learning for Anyone, Anywhere

Studynotion partners with more than 275+ leading universities and companies to bring flexible, affordable, job-relevant online learning to individuals and organizations worldwide.

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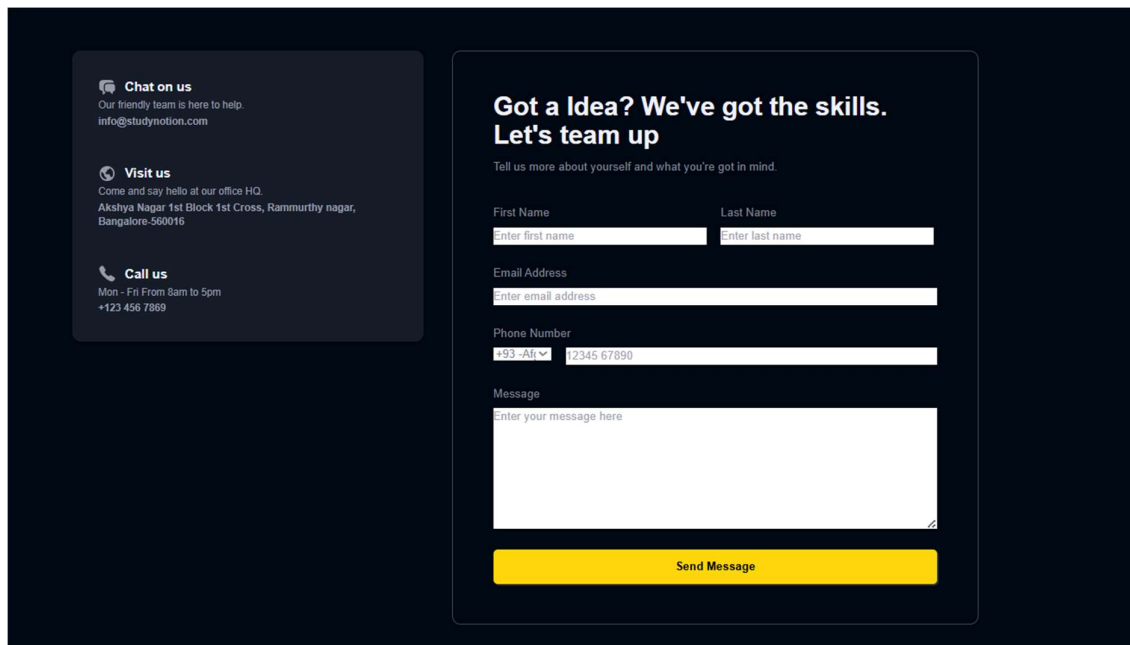
Message

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Reviews from other learners

Fig 3.11 About Us Page



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Fig 3.12 Contact Us Page

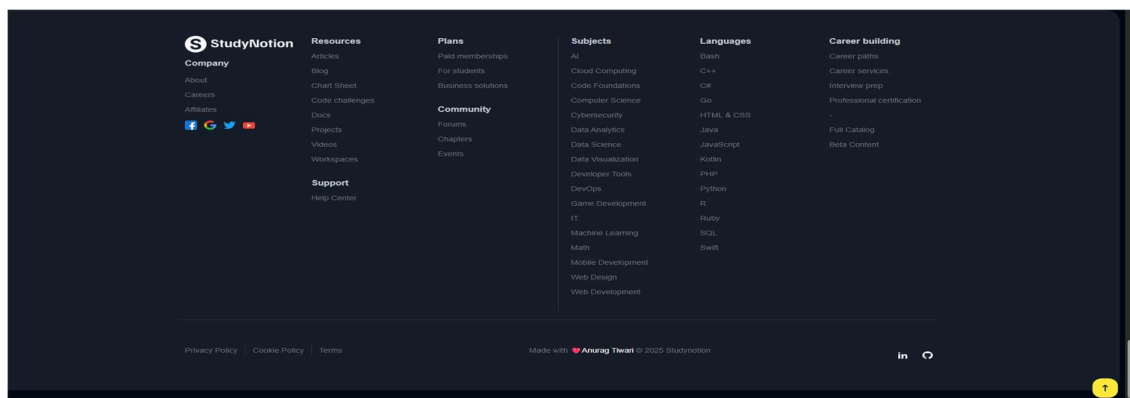


Fig 3.13 Footer Page

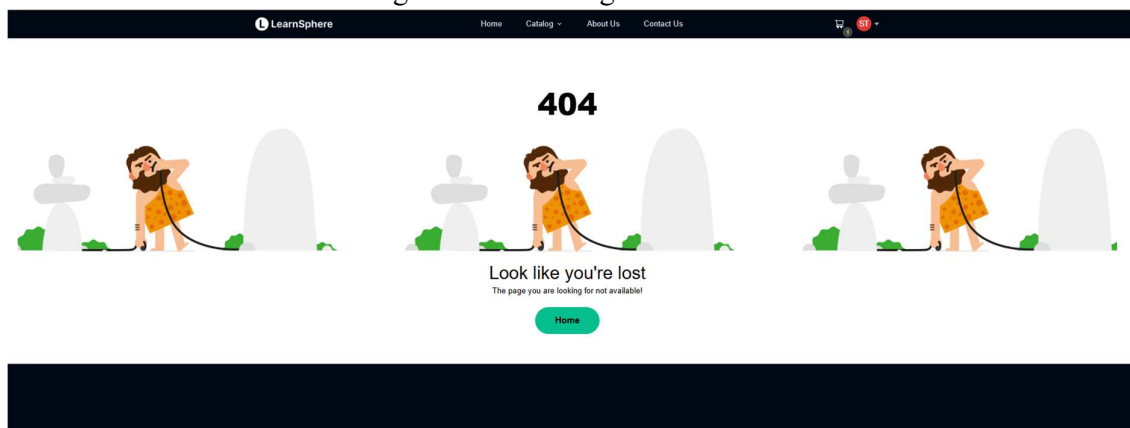


Fig 3.14 404 Not Found Page

User Management	Handles user registration, login, profile updates, and role-based (Instructor, Admin) using JWT authentication.
Course Management	Instructors can create, edit, or delete courses; manage sections, lectures, videos/documents, quizzes, and pricing.
Learning Module	Enables students to enroll in courses, access content (videos, PDFs, quizzes), track progress, and earn certificates.
Database	Central storage for users, courses, enrollments, payments, reviews, and analytics.
Admin Panel	Admin dashboard for managing users, instructors, courses, payments, and generating reports/analytics.
Payment & Checkout	Manages cart, wishlist, secure payments (Razorpay), order history, and purchase confirmations.
Security & Session	Provides JWT-based authentication, authorization, logout, and OTP verification for sensitive actions.
Search & Explore	Provides course discovery through search, filters, and categories; allows viewing instructor profiles and ratings

Table 3.1: System Modules and Their Descriptions

CHAPTER 4

DATABASE DESIGN

The database for **StudyNotion** is designed using **MongoDB**, a **NoSQL** database that stores data in flexible JSON-like documents (schemas). Each schema represents a collection of related data, and relationships are maintained via references between users, courses, enrollments, and other features.

4.1 Users Table

This schema stores information about all registered users. Fields include:

Columns include:

- **user_id:** Unique identifier for each user
- **name:** Full name of the user
- **email:** User's email (must be unique)
- **password:** Encrypted password (hashed)
- **role:** Defines user type (student, instructor, admin)
- **profile_image:** URL to profile picture (via Cloudinary)
- **bio:** Short biography or description
- **created_at:** Timestamp for account creation

4.2 Course Schema

Stores all courses created by instructors.

Fields include:

Columns include:

- **post_id:** Unique ID for each post
- **course_id:** Unique ID for each course
- **instructor_id:** References the instructor who created the course
- **title:** Course title
- **description:** Detailed description
- **price:** Course price
- **tags:** Array for search optimization
- **created_at:** Timestamp of course creation
- **updated_at:** Timestamp for last edit
-
- **thumbnail:** URL for course image (via Cloudinary)

4.3 Enrollment Schema

Tracks student enrollments in courses:

Columns include:

- **enrollment_id:** Unique identifier for each enrollment
- **course_id:** References the course
- **student_id:** References the student
- **progress:** Percentage completed
- **completed_lectures:** Array of completed lecture IDs
- **created_at:** Timestamp when enrollment occurred

4.4 Payment Schema

Stores payment transactions.

Fields include:

- **payment_id:** Unique ID for each payment
- **enrollment_id:** References the related enrollment
- **amount:** Payment amount
- **status:** (pending, completed, failed)
- **razorpay_order_id:** Reference from Razorpay
- **created_at:** Timestamp of payment

4.5 Review Schema

Manages the follow relationships between users.

Stores course reviews and ratings.

Fields include:

- **created_at:** Timestamp when the follow occurred
- **review_id:** Unique ID for each review
- **student_id:** References the student
- **rating:** Numeric rating (1-5)
- **comment:** Text review

Table 4.1: Database Design

Title	Description
Users Collection	Stores user details: user ID, name, email, password, role (Student/Instructor/Admin), profile image, bio, created_at.
Courses Collection	Contains course details: course ID, instructor reference, title, description, price, thumbnail, sections, tags, created_at, updated_at.
Enrollments Collection	Tracks student enrollments: enrollment ID, course reference, student reference, progress, completed_lectures, created_at.
Payments Collection	Stores payment information: payment ID, user reference, course reference, amount, status, razorpay_order_id, created_at.
Notifications Collection	Tracks system notifications: notification ID, user reference, type, reference ID, read status, created_at.
Reviews Collection	Stores course reviews: review ID, course reference, student reference, rating, comment, created_at.
Messages Collection	Stores direct messages: message ID, sender reference, receiver reference, content, media_url, read_status, created_at.

CHAPTER 5

SECURITY & SYSTEM FEATURES

StudyNotion is designed to be simple, secure, and user-friendly. It provides useful features for students, instructors, and administrators. The system is built with security in to ensure that only valid users can access the platform, interact safely, and that user data remains protected. This chapter explains both the main system features and the security mechanisms that safeguard the platform and its data.

This chapter explains both the **main system features** and the **security mechanisms** that safeguard the platform and its data.

5.1.1 Authentication & Authorization

Authentication ensures that users prove their identity (via login).

Authorization ensures that authenticated users have the correct permissions.

JWT (JSON Web Tokens) are used in the backend to authenticate users securely and

5.1.2 Password Security

User passwords are never stored in plain text. Instead:

- **bcrypt** is used to hash and salt passwords.
- During login, the hashed version of the entered password is compared.
- This prevents brute-force and rainbow table attacks.

5.1.3 Admin Panel

A secured section of the dashboard accessible only to admin users.

Allows:

- Managing users (promoting, disabling)
 - Moderating posts and comments
 - Viewing and updating Coourses
 - Monitoring platform statistics (analytics, activity trends)
- Protected using role-based access control.

5.1.4 Secure Messaging

Direct messages between users are private and secure:

- Encrypted communication ensures only sender and receiver can access content.
- Attachments are securely stored with access tokens via Cloudinary .
- Read/unread status is tracked without exposing sensitive info.

5.1.5 Responsive Design

The platform is mobile-friendly using modern frontend frameworks.

- Automatically adjusts layout, text, and elements for different screen sizes.
- Ensures excellent UX across desktops, tablets, and smartphones.

5.1.6 Secure Data Storage

All sensitive data (users, posts, messages, notifications) is stored securely in PostgreSQL (Supabase).

- **HTTPS** is enforced for secure communication.
- **Tokens** verify user requests.
- Prevents data breaches and tampering.

5.1.7 Backend Data Validation

All incoming data from users is validated and sanitized at the backend.

- Inputs such as login, posts, and comments are checked and cleaned.
- Middleware ensures no invalid or malicious data enters the database.
- Protects against SQL/NoSQL injection and XSS attacks.

5.1.8 Content Filtering & Search

Posts can be searched and filtered.

- Search by hashtags, usernames, or keywords.
- Filter posts by privacy settings (public, followers, private).
- Improves content discoverability and user engagement.

5.1.9 Enrollment & Notification Management

- Students can enroll securely via payments.
- Real-time notifications for enrollments, course updates, and progress.
- Notifications include read/unread tracking

5.1.10 Admin Dashboard Features

The admin dashboard provides insights and controls

- Total number of users, courses, and enrollments
- Trending courses and top instructors.
- Reports of issues or low-performing content

Authentication & Authorization	Secure login and signup using JWT with role-based access control (Student, Instructor, Admin).
Password Security	Passwords are hashed and stored securely using bcrypt with salting
User Roles	Role-specific permissions: Students (consume content), Instructors (create and manage courses), Admins (monitor and control platform).
Course-Management	Instructors can create, update, and delete courses with structured sections, lectures, videos, and thumbnails.
Responsive Design	Ensures the platform adapts seamlessly across all devices and screen sizes.
Secure Data Storage	User data, posts, and messages are protected using encryption, tokens, and HTTPS.
Backend Data Validation	Validates server-side inputs to prevent malicious or invalid data entry.
Payment Integration	Secure Razorpay payment gateway for course purchases with order capture and verification.
Enrollment & Notification Management	Manages student enrollments securely and delivers real-time notifications.
Admin Dashboard Features	Displays platform statistics, top posts, user activity insights, and moderation tools.

Table 5.1: System Security Feature

CHAPTER 6

TESTING

Testing is a crucial part of software development. It ensures that the system works correctly, safely, and as expected. Before deploying the Social Media Dashboard, thorough testing was carried out to identify and fix errors.

The testing process included several methods to guarantee that the platform is **secure, smooth, and user-friendly**.

6.1.1 Functional Testing

Functional Testing ensures that every feature of the Social Media Dashboard works as intended.

Examples:

- Users can register, log in, and log out.
- Instructors can create, edit, and delete courses.
- Students can enroll, track progress, and receive certificates.
- Payments process correctly via Razorpay.
- Notifications are sent and received properly.

6.1.2 Security Testing

Security Testing checks the system's resistance to threats and vulnerabilities.

Examples:

- JWT token tampering attempts are blocked.
- Unauthorized access to private posts or messages is prevented.
- Passwords remain encrypted and safe.

6.1.3 Performance Testing

Performance Testing evaluates how responsive and stable the system is under different loads.

It involves:

- Checking how the dashboard behaves with multiple users online simultaneously.
- Measuring feed loading time, post rendering speed, and database query response.
- Tools like **Postman** and browser dev tools were used to simulate requests and measure results

6.1.4 Integration Testing

Integration Testing ensures that the platform's components work together seamlessly.

Examples:

- Frontend properly communicates with backend APIs.
- Database queries update posts, likes, and comments correctly.
- Notification system works alongside messaging and post interactions.

6.1.5 User Interface Testing

UI Testing ensures that the dashboard is intuitive and user-friendly.

Focus areas:

- Buttons, links, forms, and modals work as expected.
- Responsive design renders properly across desktop, tablet, and mobile devices.
- Error messages and alerts display clearly.

6.1.6 Regression Testing

Regression Testing was carried out after bug fixes and updates to ensure existing features still worked correctly.

Examples:

- New updates to messaging didn't break post functionality.
- Changes in notification handling didn't affect feed rendering.

Title	Description
Functional Testing	Verifies core features like posts, likes, comments, messaging, and follows work correctly.
Security Testing	Checks the system against unauthorized access, token tampering, and vulnerabilities.
Performance Testing	Measures system speed, stability, and responsiveness with multiple concurrent users.
Integration Testing	Ensures frontend, backend, and database modules interact correctly.
User Interface Testing	Tests usability, responsiveness, and consistency across devices.
Regression Testing	Re-tests after updates to ensure existing features remain functional.

Table 6.1: Testing Summary Table

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 Conclusion

This project demonstrates the successful development of a **full-stack EdTech LMS** called StudyNotion using the MERN stack. The system supports essential online learning features such as:

The system supports essential **LMS features** such as:

- User authentication and role management
- Course creation with media (videos, documents)
- Enrollment, progress tracking, and certificates
- Payment integration and notifications
- Instructor and admin dashboards for analytics

By leveraging **MongoDB** for flexible **storage**, **ExpressJS/NodeJS** for **APIs**, and **ReactJS** with **Redux/TailwindCSS** for the **UI**, the project benefits from a scalable architecture, secure payments via **Razorpay**, and media hosting via **Cloudinary**. This makes the system both efficient to build and future-ready for deployment.

7.2 Future Scope

While the current system provides a solid foundation, several enhancements can expand its impact:

1. **Gamification** - Add badges, points, and leaderboards for student motivation..
2. **Mobile App: Develop a React Native app for iOS/Android.**
3. **AI Recommendations: Use ML for personalized course suggestions. Push Notifications**
4. **VR/AR Integration:** Support immersive learning experiences.
5. **Advanced Analytics:** Real-time dashboards with predictive insights.
6. **Internationalization:** Multi-language support.
7. **Community Features:** Forums and group discussions.
8. **Offline Access:** Sync course content for offline viewing.

Appendix A: Software and Tools Used

Tool / Technology	Purpose
ReactJS (with Redux , Tailwind-CSS)	Frontend UI and state management
ExpressJS & NodeJS	Backend APIs and server logic.
MongoDB (with Mongoose	Database for data storage
Razorpay	Payment gateway integration.
Cloudinary	Media hosting (images, videos, documents)
Visual Studio Code	Code editor
Postman	API testing
Git & GitHub	Git & GitHub Version control and collaboration

Appendix B: System Requirements

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 10GB free space
- Internet: Required for hosting and real-time features

Software Requirements:

- Operating System: Windows / Linux / macOS
- Node.js and npm installed
- Expo CLI (for React Native development)
- Supabase account (with project & database setup)
- Android Studio / Xcode (for mobile testing)
- Any modern browser (for web app testing)

Appendix C: Sample Data (**MongoDB Schemas**)

- **users** – stores user details (id, name, email, password hash, bio, profile_pic)
- **courses** – stores courses with title, description, price, sections, lectures, thumbnail
- **enrollments** – stores enrollments with progress and completed_lectures
- **payments** – stores payment details with status and razorpay refs
- **reviews** – stores course ratings and comments
- **notifications** – stores system notifications (like, comment, follow)

Appendix D: Frontend Routes (React Screens)

- /login – User login page
- /register – New user registration
- /courses – Course catalog
- /course/:id – Course details and enrollment
- /dashboard/student – Student progress dashboard
- /dashboard/instructor – Instructor course management
- /settings – User account settings
- /about – App details
- /admin – Admin dashboard (manage users, courses, reports)
-

Appendix E: REST API Design

- Sample Endpoints:
- POST /api/auth/register – Register a new user
- POST /api/auth/login – User login with JWT
- GET /api/courses – Fetch course catalog
- POST /api/courses – Create a new course (instructor only)
- POST /api/enrollments – Enroll in a course
- POST /api/payments – Initiate payment via Razorpay
- GET /api/analytics/:instructorId – Fetch instructor analytics
- GET /api/admin/users – Fetch all users (admin only)

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- [2] Express.js Documentation. Available: <https://express.com>
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- [4] *React Navigation*. Available: <https://reactnavigation.org/docs>
- [5] *ShadCN UI*. Available: <https://ui.shadcn.com>
- [6] *Radix UI*. Available: <https://www.radix-ui.com>
- [7] MERN Stack EdTech Project (GitHub Repository) Available:
<https://github.com/anuragtiwari3100/Edtech-MERN-Stack-Online-Learning-Platform>



Marwadi Chanderana Group

NAAC



Department of Computer Engineering

Marwadi University

Academic Year: 2025-26

Semester: 7

Major Project-I (01CE0716)

Weekly Progress Report Diary (Project)

Team ID: 7CE-158

Project Title: Online Learning platform.

Sr. No.	Student Full Name	Student En. No.	Class
1	ANURAG TIWARI	92201703100	EC1
2			
3			

Internal Guide Name:



Weekly Project Progress Report Diary – June

Week	Project Activity by Students	Updates / Comments / Suggestions / Remarks by Faculty	Date & Time	Guide Signature
1	Project assigned and initial understanding completed.	Project assigned	31/5/2025	
2	Completed Project Setup, including required dependencies setting up optimal folder structure.	Done	6/6/2025	
3	Completed creation of all required data model & up for project need.	Done	13/6/2025	
4	Implemented core backend APIs & Used Authentication and Authorization functionality.	Done	23/6/2025	



Weekly Project Report Diary – July

Week	Project Activity by Students	Updates / Comments / Suggestions / Remarks by Faculty	Date & Time	Guide Signature
1	Integrated No-demanded for Email services & password reset functionality.		30/6/2025	
2	Integrated cloudfog for file upload replaced tag schema with category schema		7/7/2025	
3	Integrated razorpay for handling course including setup and configuration.		16/7/2025	
4	Added category page details to fetch course by category.		24/7/2025	



Weekly Project Report Diary – August

Week	Project Activity by Students	Updates / Comments / Suggestions / Remarks by Faculty	Date & Time	Guide Signature
1	Added Rating and Review page as per chosen course.		4/08/25	RS
2	Create structured format for front end part of our project.		11/08/25	RS
3	implemented course catogry dropdown to filter courses by category eg. devops, AI	Done		RS
4	course selection & purpose flow to select a course from chooge catogry	Done	30/08/25	RS

REVIEW CARD: Term Work: Regular Reporting & Project Report

Team ID:	Team Size: <u>1</u> Project ☒ / Internship ☐	Date:
Student 1 Enrollment No.	<u>92201703100</u>	Student 1 Class: <u>EC-3</u>
Student 1 Name	<u>Amurug tikun</u>	Student 2 Class:
Student 2 Enrollment No.		Student 3 Class:
Student 2 Name		
Student 3 Enrollment No.		
Student 3 Name		
Project/Internship Title	<u>Online Learning with Meta Stack</u>	
Internal Guide Name	<u>Prof. Rakesh Kumbhar</u>	
External Guide Name (If Internship)		
Internship Company (If Internship)		

Project Report Evaluation (Total 20 marks)

Tasks (10 Marks for each task)	Internal Guide Remarks	Assigned Marks (Total 20 marks) (common for all students)
Report Format as per Guidelines	<u>Yes - Done</u>	<u>10</u>
Organization of Chapters, Description of Concepts, Tasks and Technical Details along with all necessary diagrams/charts /graphs, Citations of adequate references	<u>everything Included</u>	<u>10</u>

Regular Reporting Performance Evaluation (Total 10 marks)

Tasks (05 Marks for each task)	Internal Guide Remarks	Assigned Marks (Total 10 marks) (individual student performance)		
		Student 1	Student 2	Student 3
Project/Internship: Reporting Regularity to Internal/External Guides	<u>Regular</u>	<u>5</u>		
Project: Performance & Individual Contribution in Team Internship: Attendance History / Record (on official letter head duly stamped and signed by HR Team)	<u>Perform good</u>	<u>5</u>		

Student 1 Signature:

Student 2 Signature:

Student 3 Signature:

Internal Guide Signature:

HoD Signature

REVIEW CARD: REVIEW 1

Team ID: 7CE-158	Team Size: 1	Project	Date:
Student Name	Anurag Tiwari		
Enrollment No.	92201703100		
Class	7ECL		
Internal Guide Name	Prof Rupesh Kangiya		
Title of Project	Online Learning Platform.		

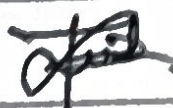
Performance Evaluation (Out of 20 marks)

Task	Reviewer 1 Evaluation Remarks	Reviewer 2 Evaluation Remarks	Assigned Marks
Introduction and Abstract (02 marks)	Explained clearly.	known about project	02
Literature Review/Survey of Existing latest systems/Business Models/Products (05 marks)	Done	Mentioned in Purpose	02
Tools & Technology, Proposed Approach / Solution / Methodology (04 marks)	Aware about tool & technology.	used Java Script, Node.js, cloud platform.	04
Implementation/Execution Flow with Planning and Scheduling, Expected Outcome (05 marks)	Basic implementation done. (UI).	Backend Model is ready	04
Presentation and Q&A (04 marks)	Satisfactory answers given.	yes	03

Suggestions by Panel Members

Panel Members	Improvement Comments/Suggestions
Reviewer 1	Need to work on backend
Reviewer 2	Work on UI/UX part on Backend & Also start work on Front End Part.
Internal Guide	Perform all the task Assigned by Reviewers

Panel Members

Panel Members	Faculty Name	Signature
Reviewer 1	Dr. Megha Madholkar	
Reviewer 2	Prof. Pooja Parmar	
Internal Guide	Prof. Rupesh Kanojia	

Anurag Tiwari

Student Signature



HoD Signature

REVIEW CARD: REVIEW 2

Team ID:	Team Size:	Project	Date:
Student Name	ANURAG TIWARI		
Enrollment No.	9229703100		
Class	7EC3		
Internal Guide Name	Prof - Purnesh Kanoria		
Title of Project	online learning platform		

Performance Evaluation (Out of 20 marks) (should be decided mutually by both reviewers)

Task	Reviewer 1 Evaluation Remarks	Reviewer 2 Evaluation Remarks	Assigned Marks
Summary & Evaluators' Remarks of Review 1 (02 marks)	Completed	completed	02
Changes/updates done based on Review 1 comments/remarks (04 marks)	Modification in ppt & ppt diagrams.	Modification in ppt & add diagram.	03
Implementation (100%) of proposed work (10 marks)	Full Module working & complete IT.	Complete all modules	08
Presentation and Q&A (04 marks)	Satisfied.	satisfied	03

Suggestions by Panel Members

Improvement Comments/Suggestions from Panel Members
Complete implementation & submit project report.

Panel Members

Panel Members	Faculty Name	Signature
Reviewer 1	Prof. Parth Parmar	
Reviewer 2	Dr. Megha Mudholkar	

Anurag Tiwari

Student Signature

HoD Signature

Follow Up Remarks by Internal Guide after Review 2
for the final Review Prepare Report & Submitme
Internal Guide Name & Signature: Purnesh Kanoria

REVIEW CARD: VIVA

Team ID: <u>7CE-158</u>	Team Size: <u>1</u>	Project	Date: <u>30/08/2025</u>
Student Name	<u>Anurag Tiwari</u>		
Enrollment No.	<u>92201703100</u>		
Class	<u>7EC-3</u>		
Internal Guide Name	<u>Prof. Rupesh Kanojiya</u>		
Project Title	<u>LMS</u>		

Performance Evaluation (Out of 30 marks) (should be decided mutually by both reviewers)

Task	Reviewer 1 Evaluation Remarks	Reviewer 2 Evaluation Remarks	Assigned Marks (30 marks)
Introduction to Project, Individual/Team Roles & Responsibilities (5 Marks)	<u>Working alone & all work complete & satisfied</u>	<u>Working alone & satisfied with performance</u>	<u>04</u>
Tools & Technologies learned (10 marks)	<u>Satisfied</u>	<u>Satisfied</u>	<u>08</u>
Project Implementation, Status of Completion, Status of Outcome 1) Research paper 2) POC 3) Start Up (Patent IDF draft is mandatory for all) (10 marks)	<u>Need to Add Some Novelty</u>	<u>Need to add some novelty</u>	<u>07</u>
Presentation and Q&A (05 marks)	<u>Satisfied</u>	<u>Satisfied</u>	<u>04</u>

Suggestions by Panel Members

Improvement Comments/Suggestions from Panel Members
<u>Add some novelty work in the IDF form & go for the further process.</u>

Panel Members

Panel Members	Faculty Name	Signature
Reviewer 1	<u>Prof. Parth Parmar</u>	<u>[Signature]</u>
Reviewer 2	<u>Dr. Megha Mudholkar</u>	<u>[Signature]</u>

Anurag Tiwari

Student Signature

HoD Signature

Remarks by Internal Guide
<u>Please Submit final Report, Share Deploy project link and github link as well</u>
Internal Guide Name & Signature: <u>Rupesh Kanojiya</u> <u>[Signature]</u>